
Chronos T5 Tiny vs. Classical Forecasting Methods

An applied foundation-model comparison across six datasets,
nine models, and five metrics

Foundation: Chronos T5 Tiny

Classical: seven methods

Neural reference: DeepAR

Prepared by

Amirhossein Bagheri

Politecnico di Milano

Supervisor: Tommaso Bianchi

What the project compares

Evaluate whether Chronos T5 Tiny, a pretrained **foundation time-series model**, can match or outperform **classical forecasting methods** under one common pipeline. DeepAR remains a separate neural reference.

Accuracy comparison

Which model minimizes median-forecast error?

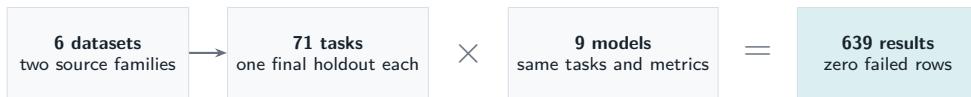
Uncertainty comparison

Which model produces the best quantile forecasts?

Practical tradeoffs

Is the comparison valid, and what does each method cost to run?

Historical snapshot: six datasets and nine models



Dataset	Tasks	Horizon	Lag
M1 yearly	20	6	1
M4 hourly	20	48	24
Weather	20	30	7
National illness	1	24	52
Traffic	5	24	24
PSM	5	24	24

Model families

- ▶ **Foundation model:** Chronos T5 Tiny
- ▶ **Classical methods:** seasonal naive, AR, MA, ARMA, ARIMA, AutoARIMA, Prophet
- ▶ **Additional neural reference:** DeepAR

Historical single-origin run. The repaired full rerun is pending.

Read the metric before reading the ranking

Point forecast metrics

MAE	Original units; easy to explain, scale-dependent.
RMSE	Original units; emphasizes large misses.
SMAPE	Range 0–2 here; unstable around zero.
MASE	Relative to an in-sample seasonal-naive scale.

MASE < 1: below the in-sample naive error scale

Probabilistic metric

WQL averages pinball loss across quantiles $0.1, \dots, 0.9$ and normalizes by total absolute target.

- ▶ Dimensionless and unbounded above.
- ▶ Official aggregation weights by target magnitude.
- ▶ Chronos/DeepAR use samples; ARIMA uses Gaussian intervals.
- ▶ Seasonal naive repeats one point at every quantile.

All five metrics are lower-is-better, but they answer different questions.

Historical result: no single winner across all tasks

Dataset	Point result	WQL winner	Main caveat
M1 yearly	DeepAR raw error; ARIMA scaled error	DeepAR	Largest series dominates
M4 hourly	Seasonal naive raw; Chronos scaled	Chronos	Objective changes winner
Weather	No clean winner	Chronos, narrowly	Zero-collapse confirmed
National illness	ARIMA(2,1,2)	Prophet	Historical: one cutoff
PSM	ARIMA(2,1,2)	Chronos	Historical time-scale error
Traffic	DeepAR	DeepAR	First five sensors only

Best classical reference

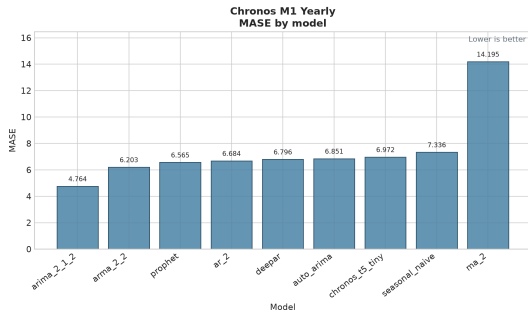
ARIMA(2,1,2), mean rank 2.86 after excluding invalid weather SMAPE.

Amirhossein Bagheri

Foundation-model result

Chronos is strong on M4 and selected WQL comparisons, but not everywhere.

M1 yearly: winner depends on how series are weighted



Raw-scale winner

DeepAR: MAE **445,885**, RMSE **543,050**,
official WQL **0.193**.

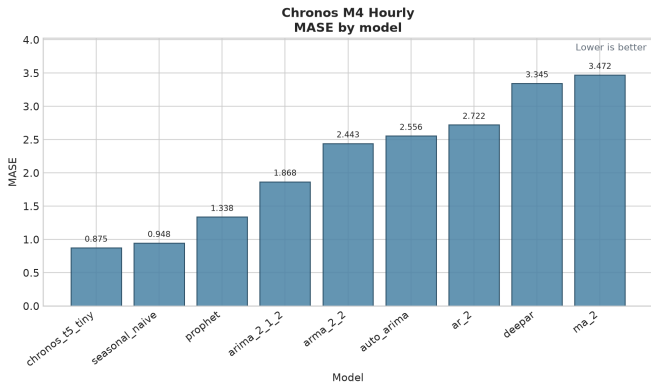
Scaled-error winner

ARIMA(2,1,2): SMAPE **0.216**, MASE **4.764**.

Interpret carefully

- ▶ Largest series carries 76.1% of WQL target weight.
- ▶ Every model has average MASE above 1.
- ▶ MA(2) WQL 131.1 is a variance explosion.

M4 hourly: Chronos and seasonal naive separate the field



Seasonal naive

Best MAE **473.3** and RMSE **590.7**;
total runtime about **0.01 s**.

Chronos

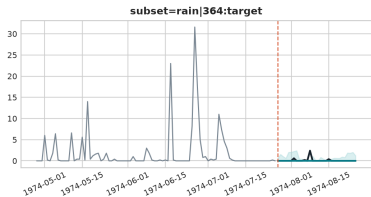
Best SMAPE **0.0452**, MASE **0.875**,
and official WQL **0.0288**.

- ▶ Chronos wins WQL on 15 of 20 series.
- ▶ Both leaders have MASE below 1.
- ▶ Prophet is a distant but consistent third.

Weather: the apparent winner predicts almost exactly zero

Chronos Weather | chronos_t5_tiny Forecast versus actual (latest rolling origin)

— History — Actual — Point forecast 10-90% interval



Historical pathology

Chronos SMAPE is exactly **2.0** for all 20 series. Reproduced median predictions are only 1.3×10^{-8} to 4.6×10^{-8} .

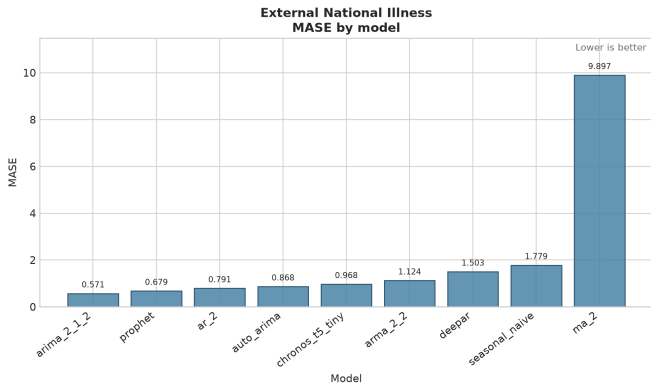
Corrected visual audit

The representative rainfall holdout is 86.7% zero; Chronos predicts zero at every step.

- ▶ Audit WQL: Chronos 1.146 vs. zero baseline 1.000.
- ▶ Occurrence error and positive-only MAE are now saved.
- ▶ Weather is explicitly filtered to the rain subset.

Conclusion: no defensible overall weather winner yet.

National illness: classical models lead one test case



Point forecasts

ARIMA(2,1,2) wins MAE, RMSE, SMAPE, and MASE. Its MASE is **0.571**.

Quantile forecasts

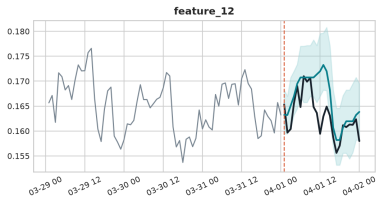
Prophet WQL is **0.0304**, about 9.1% below ARIMA's **0.0334**.

- ▶ Chronos ranks fifth by MASE: 0.968.
- ▶ MA(2) is last by a wide margin.
- ▶ Historical result used one OT cutoff.
- ▶ Repaired configuration evaluates three rolling origins.

PSM: historical ranking invalid; time scale repaired

External Psm | chronos_t5_tiny
Forecast versus actual (latest rolling origin)

— History — Actual — Point forecast 10-90% interval



Do not select a PSM model from the historical scores; rerun the repaired benchmark.

Historical ranking

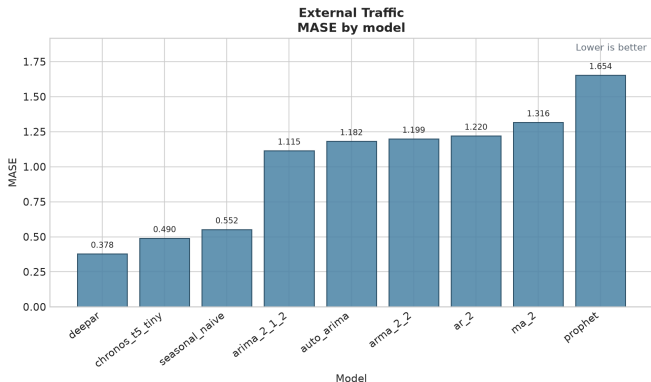
ARIMA(2,1,2) wins all point metrics: MASE **0.270**. Chronos wins official WQL: **0.00375**.

Implemented repair

Source minutes are parsed and resampled to hourly means; horizon 24 and lag 24 now both mean 24 hours.

- ▶ Audit trajectory uses genuine hourly timestamps.
- ▶ Features are sampled evenly instead of taking the first five.
- ▶ Historical scores still require a full rerun.

Traffic: DeepAR wins; Chronos remains competitive



DeepAR

Wins MAE **0.00765**, RMSE **0.01057**, MASE **0.378**, and WQL **0.1466**.

Alternatives

Chronos wins SMAPE at **0.198**; seasonal naive has MASE **0.552** at essentially zero runtime.

- ▶ DeepAR wins four of five sensors.
- ▶ Top three models all have MASE below 1.
- ▶ Prophet is last on every metric.

Scope: the first five columns out of 862 traffic signals.

Historical scorecard: ARIMA first, Chronos second

Model	Mean rank*	Observed strength	Observed weakness
ARIMA(2,1,2)	2.86	Illness/PSM point; M1 scaled	M4, weather, traffic
Chronos T5 Tiny	3.52	M4; PSM WQL; traffic	M1; zero-like weather median
DeepAR	4.62	M1 weighted; traffic	M4 and PSM
Prophet	4.69	M1; illness WQL	PSM and traffic
Seasonal naive	4.72	Speed; M4; strong baseline	No calibrated uncertainty
AutoARIMA	4.79	Illness/PSM point	No metric wins; costly search
ARMA(2,2)	4.90	M1/PSM scaled error	Slow; no metric wins
AR(2)	6.07	Simple illness baseline	Mostly lower-middle
MA(2)	8.83	None demonstrated	Weak and unstable uncertainty

*Mean within-dataset rank over all metric cells except invalid weather SMAPE; descriptive project summary, with correlated metrics and unequal sample sizes.

What is historical, and what is now repaired

Historical result snapshot

- ▶ No model dominates every dataset and objective.
- ▶ ARIMA(2,1,2) is the strongest classical reference.
- ▶ Chronos is strong on M4 and probabilistic PSM scoring.
- ▶ DeepAR is strongest on the tested traffic sensors.
- ▶ Seasonal naive remains difficult to beat per unit runtime.

Repaired evaluation pipeline

- ▶ Three rolling origins per selected series.
- ▶ Three seeded runs for stochastic models.
- ▶ Evenly spaced representative sampling.
- ▶ Hourly PSM and rainfall-specific diagnostics.
- ▶ Forecast values and 55 trajectory plots persisted.

The full repaired rerun is required before updating model rankings.

How to choose a model, and what to fix next

Speed / baseline

Seasonal naive

Point accuracy

ARIMA(2,1,2)

Probabilistic quality

Compare WQL by dataset

Weather or PSM?

Rerun before selection

Completed repairs

- ▶ Correct PSM timestamps and seasonal lag.
- ▶ Add zero and occurrence baselines for rainfall.
- ▶ Align WQL plotting with official aggregation.
- ▶ Persist forecast trajectories for every model.

Next validation step

- ▶ Run all three origins and stochastic seeds.
- ▶ Regenerate metric tables and this presentation.
- ▶ Report dispersion and interval calibration.

Chronos expands the toolkit; classical methods remain essential.

Use validated metrics to choose by dataset,
and keep strong classical baselines in every comparison.

Metric-aware

Baseline-grounded

Validity-checked